

ECON 4720: HA03

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Problem 1

Are the following vectors linearly independent?

Problem 1a

$$a_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 7 \\ 6 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 12 \\ 10 \end{pmatrix}$$

Problem 1b

$$a_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Problem 2

Find determinants of the following matrices:

$$A = \begin{bmatrix} 1 & 5 & 7 \\ 3 & 2 & 8 \\ 6 & 1 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 & 2 \\ 6 & -2 & 3 \\ 7 & 10 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 13 & 4 & 1 \\ 0 & 4 & 1 \\ -7 & 2 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} -4 & 4 & 12 \\ 3 & -3 & -9 \\ 8 & 2 & 6 \end{bmatrix}$$

Problem 3

Using definitions, check whether the following matrices are positive definite or positive semidefinite:

Problem 3a

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}, \quad E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Problem 3b

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

Problem 4

Find the rank of each the following matrices:

$$A = \begin{bmatrix} 3 & 6 & 4 & 8 \\ 2 & 7 & 1 & 9 \\ 4 & 2 & 5 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 18 & 2 \\ 7 & -4 \\ 6 & 11 \end{bmatrix}, \quad C = \begin{bmatrix} -5 & 8 & 11 & 0 \\ 13 & 3 & 0 & 2 \\ 10 & 0 & -6 & 2 \end{bmatrix}$$

Problem 5

Find such value of the parameter α that the following vectors are linearly dependent:

$$a_1 = \begin{pmatrix} 2 \\ \alpha \\ -2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}, \quad a_3 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

Problem 6

Solve the following systems of linear questions:

Problem 6a

$$\begin{cases} x + y + z = 11 \\ x - y - z = -3 \\ -x + y - z = 5 \\ 3x - y + 2z = 2 \end{cases}$$

Problem 6b

$$\begin{cases} x + y + z + t + p = 17 \\ x - y - z - t - p = -5 \\ z + t + p + y = 11 \\ p - x - y = 1 \\ -t + x = 10 \end{cases}$$

Problem 6c

$$\begin{cases} x + y + z + t = -6 \\ x - y - z - t = 20 \\ y - x = -39 \\ 2x + 3t + y - 4z = 32 \end{cases}$$

Problem 6d

$$\begin{cases} -4x - 5y - 5z = -74 \\ -3x - 5y - z = -4 \\ -x + 7y + 6z = 48 \end{cases}$$

Problem 6e

$$\begin{cases} x + 2y - z + 3t = 0 \\ x + 2y + 2z - t = 1 \\ 3x + 6y + 5t = 1 \\ 4x + 8y - z + 8t = 1 \end{cases}$$

Problem 7

For which values of λ the following matrix is invertable?

$$A = \begin{bmatrix} 1 & -2 & 2 \\ \lambda & 3 & 0 \\ 2 & 1 & 1 \end{bmatrix}$$

Problem 8

Find all values a , for which the system of linear equations does not have any solution.

$$\begin{cases} 2x + (9a^2 - 2)y = 3a \\ x + y = 1 \end{cases}$$

Problem 9

Find a value of b such that system of linear questions has infinitely many solutions:

$$\begin{cases} 6x + 2by = 3 \\ 4x + 2y = 2 \end{cases}$$

Problem 10

For the following matrices check the statement: “If a matrix X has full rank then the matrix $X'X$ is invertible” and find the inverse $(X'X)^{-1}$:

Problem 10a

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Problem 10b

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

Problem 10c

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 0 \\ 4 & 1 \end{bmatrix}$$